

S DURGA PRASAD

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PROFESSIONAL SUMMARY

Software Engineering student with expertise in **C++ Backend Development, Networking, AI/ML Applications, and Software Engineering**, experienced in designing and developing scalable applications using **C++, Java, Python, SQL, REST APIs, Docker, Linux, and Cloud Technologies**. Skilled in **backend systems, distributed applications, network automation, network diagnostics, deployment workflows, database design, debugging, testing, and product development**. Hands-on experience in building **AI/ML solutions, automation pipelines, intelligent data-processing systems, authentication workflows, and cloud-native applications**. Strong understanding of **Computer Networks, TCP/IP, Operating Systems, Data Structures, Algorithms, SDLC, Agile Methodologies, and Software Architecture**. Proven ability to solve complex technical problems, collaborate in cross-functional teams, and contribute to end-to-end product development in fast-paced technology environments.

EDUCATION

Vellore Institute of Technology

M.Tech (Integrated) Software Engineering — CGPA: 8.62

Vellore, India

Sept 2022 – Present

PROJECTS

Hand Biometric Authentication System — Python, TensorFlow, OpenCV, Streamlit

- Developed secure biometric authentication system using Python, Deep Learning, and Computer Vision techniques
- Implemented authentication workflows, intelligent validation systems, and secure backend communication pipelines
- Integrated AI/ML models, scalable processing systems, and Linux-compatible deployment workflows
- Applied Cyber Security concepts, secure APIs, encryption workflows, and intelligent automation systems
- Improved system reliability through debugging, testing, optimization, and scalable architecture implementation

End-to-End ML Pipeline System — Python, Docker, MLFlow, CI/CD

- Developed scalable ML pipeline processing 5000+ records using Python, Docker, and automation workflows
- Implemented AI technologies, backend automation systems, distributed workflows, and cloud-native deployment
- Reduced manual effort by 40% through workflow automation, scalable infrastructure, and intelligent orchestration
- Worked with Linux environments, debugging, testing, monitoring systems, and CI/CD pipelines
- Applied Machine Learning concepts, automation systems, and scalable application development methodologies

Smart Farming Platform — Python, PostgreSQL, REST APIs

- Developed backend application managing 1000+ records using Python, REST APIs, and scalable system architecture
- Implemented automation workflows, database integration, monitoring systems, and Linux-based backend services
- Enhanced operational efficiency by 25% through debugging, optimization, testing, and modular development
- Worked with distributed systems, backend APIs, scalable deployment workflows, and intelligent processing systems
- Applied SDLC principles, Agile methodologies, and secure application development practices

NoteShare – Student Collaboration Platform — Java, SQL, Backend Systems

- Built scalable collaboration platform used by 2000+ users with secure backend systems and database integration
- Implemented authentication systems, secure workflows, REST APIs, and distributed backend communication
- Handled 500+ user requests through optimized SQL processing, scalable services, and modular architecture
- Applied Cyber Security basics, access control systems, debugging, testing, and workflow optimization
- Worked with scalable deployment workflows, Linux-compatible systems, and backend application development

TRAINING & PROGRAMS

NxtWave CCBP 4.0 – Industry Ready Software Development Program

Sept 2023 – Present

- Training in Python, Java, C++, Data Structures, Algorithms, Backend Development, and Linux systems
- Hands-on experience in secure application development, REST APIs, automation workflows, and scalable systems
- Worked with AI/ML workflows, debugging, testing, CI/CD pipelines, and cloud-native deployment concepts
- Exposure to Cyber Security fundamentals, Operating Systems, networking concepts, and distributed systems
- Practiced Agile methodologies, workflow optimization, system integration, and intelligent automation development

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, SQL

Cyber Security & Authentication: Secure Application Development, Authentication Systems, Access Control, Encryption Basics

Embedded & Linux Systems: Linux Operating Systems, Embedded Systems Concepts, System Integration, Automation Systems

AI & Machine Learning: Machine Learning, Deep Learning, OpenCV, TensorFlow, AI Technologies

Backend Development: REST APIs, Backend Systems, Distributed Systems, Scalable Architecture

Cloud & DevOps: Docker, CI/CD, Cloud-Native Deployment, Monitoring Systems

Databases & Tools: PostgreSQL, MySQL, Git, GitHub, MLFlow, Postman, Nmap, Wireshark, Have an overview of Kali linux tools

Software Engineering: SDLC, Agile Methodologies, Debugging, Testing, Workflow Optimization

Computer Science Fundamentals: Data Structures, Algorithms, Networking, Operating Systems

Professional Skills: Problem Solving, Communication, Collaboration, Documentation, Analytical Thinking

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